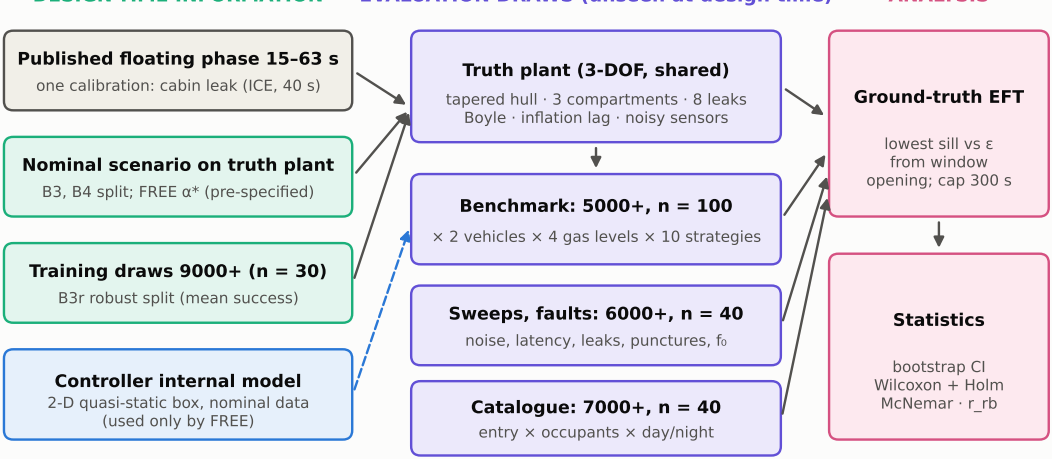
DESIGN-TIME INFORMATION
EVALUATION DRAWS (unseen at design time)
ANALYSIS
Published floating phase 15–63 s
one calibration: cabin leak (ICE, 40 s)
Nominal scenario on truth plant
B3, B4 split; FREE α* (pre-specified)
Training draws 9000+ (n = 30)
B3r robust split (mean success)
Controller internal model
2-D quasi-static box, nominal data
(used only by FREE)
Truth plant (3-DOF, shared)
tapered hull · 3 compartments · 8 leaks
Boyle · inflation lag · noisy sensors
Benchmark: 5000+, n = 100
× 2 vehicles × 4 gas levels × 10 strategies
Sweeps, faults: 6000+, n = 40
noise, latency, leaks, punctures, f₀
Catalogue: 7000+, n = 40
entry × occupants × day/night
Ground-truth EFT
lowest sill vs ε
from window
opening; cap 300 s
Statistics
bootstrap CI
Wilcoxon + Holm
McNemar · r_rb